

WORKSHEET

Ratios in other contexts

Making chocolates

Cocoa butter, cocoa liquor and sugar are the main ingredients in chocolate. Milk chocolate and white chocolate also contain milk powder. If you want to make quality chocolate, you must combine the ingredients in the correct ratios.

- 1 When Carol makes chocolate truffles she mixes melted dark chocolate and cream in the ratio of 2 : 1.
 - a How much melted dark chocolate does Carol mix with 200 mL of cream?
 - b How much cream does Carol need when she uses 500 mL of melted chocolate?
 - c Carol needs to make 1200 mL of truffle mixture. How much melted chocolate and cream will she need?
 - d Carol rolls the truffle balls in crushed nuts. She requires 15% of the volume of the truffle mixture in crushed nuts. How many mg of crushed nuts will she need for 1200 mL of truffle mixture?
- 2 Isabella is going to make some dark chocolates. She has to combine cocoa liquor, sugar and cocoa butter in the ratio of 10 : 8 : 3.
 - a What quantities of cocoa liquor and sugar will she use with 300 g of cocoa butter?
 - b Isabella only has 500 g of cocoa liquor. Calculate the quantities of cocoa butter and sugar she will need to mix with 500 g of liquor and the total weight of the resulting chocolate mixture.
 - c Isabella requires 25 g of chocolate mixture to make a heart shaped chocolate with a flavored centre. How many heart shaped chocolates will she be able to make from the mixture?

- 3** Jessica is going to make some milk chocolate for Easter eggs by mixing cocoa liquor, sugar, milk powder and cocoa butter in the ratio of 5 : 8 : 5 : 4.
- a** Of which of the four ingredients will Jessica use the largest quantity?

 - b** Which ingredient is the smallest part of milk chocolate?

 - c** Jessica is going to use 200 g of cocoa liquor. How much of the other ingredients will she need?

 - d** How many grams of sugar are in a 440 g block of milk chocolate?
- 4** Cocoa liquor is the ingredient that makes chocolate brown. White chocolate doesn't contain any cocoa liquor. It is made from cocoa butter, sugar and milk powder in the ratio of 11 : 10 : 10. Determine the quantity of each ingredient required to make 465 g of white chocolate.

Answers

- 1** a 400 mL
b 250 mL
c 800 mL melted chocolate, 400 mL cream
d 180 mg
- 2** a 1kg cocoa liquor, 800 g sugar
b 400 g sugar, 150 g cocoa butter, total weight 1050 g
c 42
- 3** a sugar
b cocoa butter
c 320 g sugar, 200 g milk powder, 160 g cocoa butter
d 160 g
- 4** 165 g cocoa butter, 150 g sugar, 150 g milk powder